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	Total No. of Page	

METHOD OF STATEMENT FOR AIR CONDITIONING & MECHANICAL VENTILATION (ACMV) SYSTEM TESTING, COMMISSIONING, AND AIR BALANCING

Prepared BY: _____
Subcontractor (Name & Signature) (Date)

Checked BY: _____
Project(M&E) (Name & Signature) (Date)
In- Charge

Approval BY: _____
Consultant (Name & Signature) (Date)

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1.0 Objective

This document outlines the systematic procedure for the testing, commissioning, and air balancing of Mechanical Ventilation Fan systems, including Fresh Air Fans (FAF) and Exhaust Air Fans (EAF). The primary goal is to verify and document that the ventilation systems are installed, operating, and performing in accordance with the project design specifications, approved shop drawings, and relevant Singapore Standards (SS 553, SS 554). This process ensures the system provides the required ventilation rates, room pressurization, and indoor air quality.

2.0 Scope of Work

This method statement applies to the complete testing and commissioning of mechanical ventilation systems, including:

- Fresh Air Fans (FAF) and Exhaust Air Fans (EAF), with or without Variable Speed Drives (VSDs) or speed regulators.
- Associated ductwork, intake/discharge louvers, supply/exhaust grilles, and dampers (VCDs, VSDs, Fire Dampers).
- Control systems, including the Local Control Panel (LCP) and interface with the Building Management System (BMS).

3.0 References

- **Workplace Safety and Health Act (WSH Act)** and subsidiary legislation.
- Project-Specific Approved Shop Drawings, Technical Specifications, and Equipment Schedules.
- Fan Manufacturer's Installation.

4.0 Responsibilities

- **Site Engineer:** Overall management and technical oversight of the commissioning process.
- **Commissioning Coordinator:** Planning, scheduling, and direct supervision of all on-site testing activities, ensuring procedures are strictly followed.
- **ACMV Technicians:** Execution of the physical testing, balancing, and adjustment works as directed.
- **Controls (BMS) Contractor:** To be present during BMS interface testing to verify signals and commands.

- **Safety Personal:** Enforcement of all HSE requirements, including review of Risk Assessments and issuance of Permits-to-Work (PTW).

5.0 Preparation

- Verify that the installation of the fan, ductwork, louvers, and grilles is 100% complete and signed off as per the approved shop drawings.
- Ensure all associated electrical works, including power supply and control wiring to the LCP, are complete and have passed initial electrical checks.
- Confirm that all required test ports for Pitot tube measurements have been installed in the main ducts at appropriate locations.
- Conduct a pre-commissioning toolbox talk with all involved personnel to review this Method Statement, the Risk Assessment, and the specific tasks for the day.

6.0 Site Equipment Performance of Air Balancing Testing & Commissioning and Hand Tools Photos

Category	Item Description
Test Equipment	1. Air Capture Hood 2. Vane Anemometer or HOT wire Anemometer 3. Multi-function Anemometer with various probes (pitot tube, temperature/humidity probe) 4. Multi Clamp Meter
Safety Equipment	1. Safety Barriers. 2. Laminated, Warning Signs. 3. Full-body harnesses and lanyards for any work at height. 4. Ladder, Scaffold, scissor lift if necessary
PPE	1. Safety Helmets. 2. Safety Goggles. 3. Safety Boots. 4. Hand Gloves.

 Air Capture Hood Instruments	 Vane Anemometer	 Multi-function Anemometer
 Multi Clamp Meter	 PPE	 Ladder

(This section in the final report would include clear, dated photographs of the specific calibrated instruments used on-site, showing their serial numbers and calibration stickers.)

7.0 Pre-Commissioning Checks

All work in this section must be performed with the fan's power source isolated and secured under a Lock-Out/Tag-Out (LOTO) procedure.

1. Mechanical Checks:

- Verify the fan model and motor details match the approved equipment submittals.
- Check for correct fan belt tension and pulley alignment.
- Manually rotate the fan impeller to ensure it moves freely without obstruction.
- Verify that all vibration isolators and flexible duct connections are correctly installed.
- Ensure all ductwork is clean and free from any construction debris.

2. Damper Checks:

- Ensure all associated Volume Balancing Dampers (VBDs) and Volume Control Dampers (VCDs) are set to the **100% open position**.
- Verify that all fire dampers are open and their fusible links are in place.

3. Electrical Checks (at LCP):

- Verify all power and control wiring terminations are tight.
- Perform insulation resistance ("Megger") tests on motor windings and power cables.
- Verify the motor overload relay is set to the correct Full Load Ampere (FLA) rating as per the motor nameplate.

9.0 Using of Equipment with Testing Method

9.1 Step-by-Step Balancing & Verification Procedure

Step 1: Initial System Start-up and Verification

1. Remove LOTO and energize the LCP.
2. Start the fan from the LCP in "Hand" mode using a brief "bump start" (on then immediately off).
3. **Verify Fan Rotation:** Check that the impeller is rotating in the correct direction as indicated by the arrow on the fan housing. If incorrect, de-energize (LOTO) and have the electrician swap two phases of the power supply.
4. Once rotation is correct, start the fan and let it run. Visually and audibly inspect for any abnormal noise, vibration, or operational issues.

Step 2: Establish Total Fan Airflow

1. Using the Multi-function Anemometer with Pitot Tube:

- At the designated test port in the main duct, perform a full Pitot tube traverse.
- Measure the velocity pressure at multiple points across the duct's cross-section to calculate the average air velocity (m/s).
- Calculate the **Total Measured Airflow** by multiplying the average velocity by the internal duct area.
- Compare this measured total with the fan's total design airflow from the equipment schedule.

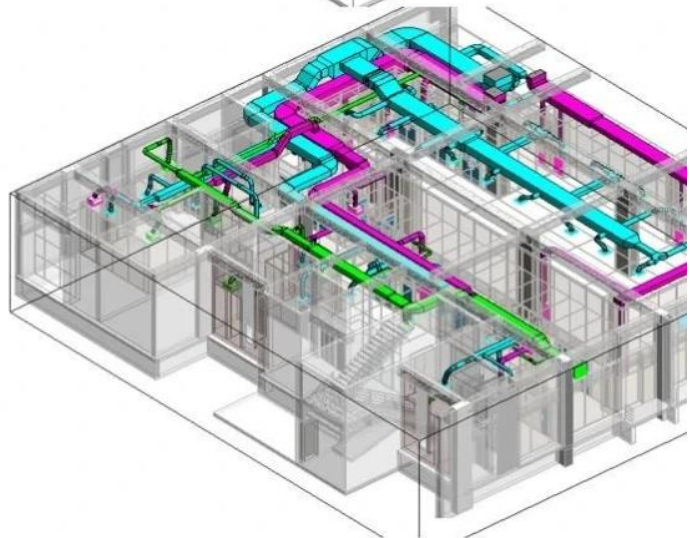
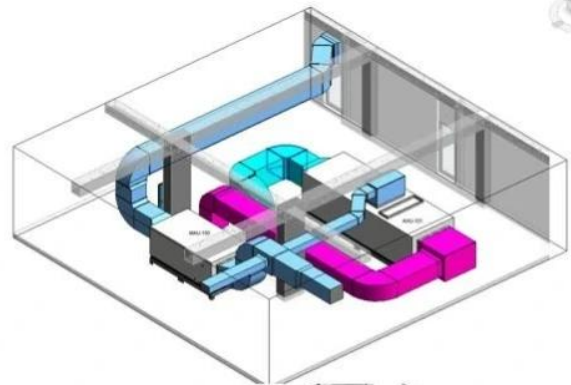
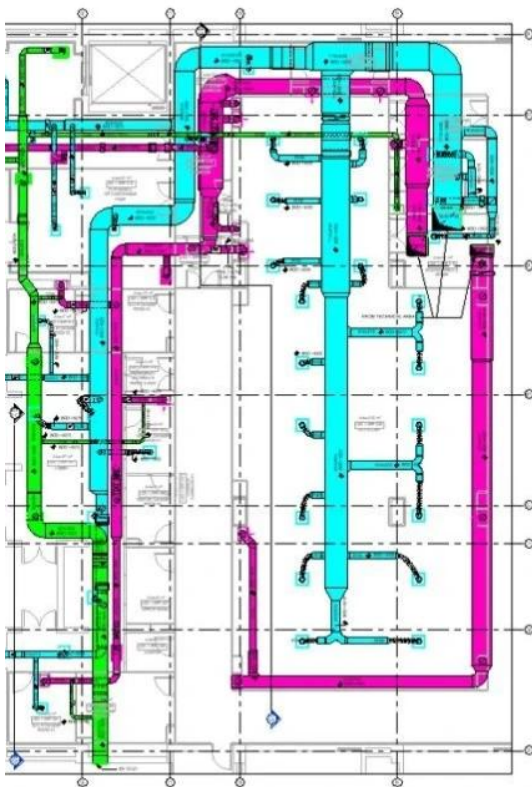
Step 3: Adjust Total Airflow (Setting the Fan)

• For Fans with a Variable Speed Drive (VSD):

1. Access the VSD controller (at the LCP or via BMS).
2. While monitoring the live airflow from the Pitot tube, adjust the VSD's operating frequency (Hz) up or down until the measured total airflow matches the design value within +/- 5%.

3. Record the final Hz and motor running amps.
- **For Fans without a VSD (Constant Speed):**
 1. If airflow is outside tolerance, perform LOTO.
 2. Adjust the fan speed by changing the motor/fan pulley sizes.
 3. Remove LOTO, restart, and re-measure. Repeat until the target airflow is achieved.

Example Plan Layout Drawings for MV System



Step 4: Proportional Balancing of the Distribution/Collection Network

1. **For FAF (Supply System):**
 - Balance the main duct branches first using their VBDs, with the "index branch" damper remaining 100% open.

- Then, balance the individual terminal supply grilles/louvers using the methods below:
- **Using the Air Capture Hood:** For standard grilles, use the hood to get a direct airflow reading. Adjust the local VCD until the design flow is met.
- **Using the Vane Anemometer:** For large louvers, perform a velocity traverse and calculate the flow. Adjust the VCD accordingly.

2. For EAF (Exhaust System):

- Go to each exhaust grille in the rooms served (e.g., toilets, kitchens).
- Using the Air Capture Hood or Vane Anemometer, measure the airflow being extracted.
- Identify the "index grille" (lowest % of design) and leave its VCD 100% open.
- Throttle the VCDs on all other grilles until each one is extracting its design airflow.

Step 5: Verify Room Pressurization

1. Using the Multi-function Anemometer (Differential Pressure Mode):

- **For EAF systems (e.g., Toilets):** The room must be under negative pressure. Measure the pressure difference between the room and the adjacent corridor (e.g., by sliding the probe under the door). Verify it meets the design requirement (e.g., -5 Pa).
- **For FAF systems (e.g., Pressurization Staircases):** The space must be under positive pressure. Verify the positive pressure relative to adjacent areas.

10.0 Using the Air Capture Hood

- **Application:** Preferred method for measuring airflow from individual air diffusers (e.g., 4-way, 3-way, 2-way) and return/exhaust air grilles due to its high accuracy.
- **Step-by-Step Procedure:**
 1. **Preparation:** Ensure the air capture hood is assembled correctly with the appropriate skirt size (e.g., 600mm x 600mm, 1200mm x 600mm) that matches the nominal size of the diffuser or grille being tested.
 2. **Power On and Zero:** Turn on the digital meter of the hood and allow it to stabilize. Perform a zero-calibration as per the manufacturer's instructions before taking the first measurement.

3. **Positioning the Hood:** Place the hood firmly against the ceiling or wall, ensuring the skirt creates a complete seal around the perimeter of the diffuser or grille. This prevents air leakage and ensures all airflow passes through the meter.
4. **Taking the Reading:** Allow the reading on the digital display to stabilize for several seconds. The meter directly displays the volumetric airflow rate in **Liters per second (L/s)** or **Cubic Meter Per Hour (m³/h)**
5. **Recording Data:** Record the measured airflow on the "Air Balancing Test Sheet" corresponding to the specific grille/diffuser tag number.
6. **Comparison and Adjustment:** Compare the measured airflow with the design airflow. If the reading is outside the acceptable tolerance (typically +/- 10% of design), locate the nearest Volume Control Damper (VCD) in the branch ductwork. Adjust the VCD lever and re-measure the airflow until the reading is within the specified tolerance.
7. **Marking Position:** Once the correct airflow is achieved, mark the final position of the VCD lever with a permanent marker to prevent future maladjustment.
8. **Repeat:** Repeat this process for all diffusers and grilles in the designated area.

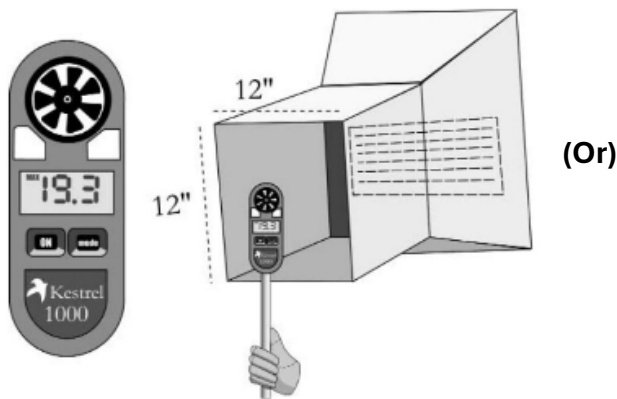


Air Capture Hood

10.1 Using the Vane Anemometer or HOT wire Anemometer

- **Application:** Used for large grilles, louvers, or where an air capture hood cannot be deployed. This method calculates airflow by measuring air velocity over a defined area.
- **Step-by-Step Procedure:**
 1. **Power On and Zero:** Turn on the digital meter of the Anemometer and allow it to stabilize. Perform a zero-calibration as per the manufacturer's instructions before taking the first measurement.
 2. **Measure Grille Area:** Accurately measure the dimensions (length and width) of the grille or louver. Calculate the "free area" based on the Approval drawing. The free area is the actual open area through which air can pass.

3. **Traverse Pattern:** Mentally divide the face of the grille into a grid of equal, smaller areas (a minimum of 9-12 readings is recommended for accuracy) or Depend on the grille size.
4. **Positioning the Anemometer:** Hold the vane anemometer head parallel to the grille face at the Center of the first grid section. Ensure the vanes are perpendicular to the direction of airflow.
5. **Take Velocity Readings:** Record the air velocity reading in meters per second (m/s). Move the anemometer to the Center of each subsequent grid section and record the velocity. This process is known as "traversing" the grille.
6. **Calculate Average Velocity:** Sum all the recorded velocity readings and divide by the number of readings taken to find the average air velocity.
7. **Calculate Airflow:** Calculate the volumetric airflow using the formula:
 - **Airflow (m³/hr) = Average Velocity (m/s) × Free Area (m²) × 3600**
8. **Recording and Adjustment:** Record the calculated airflow on the test sheet. If adjustment is needed, follow steps 6 and 7 from the Air Capture Hood procedure to adjust the VCD and re-measure.



Vane Anemometer



Air Velocity Meter

11.0 Local Control System Verification

The LCP provides the essential means to operate the equipment under different scenarios, typically managed by a **Hand-Off-Auto (H-O-A)** selector switch:

11.1 Local Control Functional Test ("Hand" Mode)

1. Turn the H-O-A selector switch on the LCP front panel to **"HAND"**.
2. Press the local **"START"** button.
 - **Verification:** The fan/motor should start and run. The **"RUN"** indicator lamp should illuminate.

3. Using a clamp-on ammeter, measure the running current on all three phases at the motor terminals or VSD output. The readings should be balanced and not exceed the motor's Full Load Ampere (FLA) rating. Record these initial readings.
4. Press the local "STOP" button.
 - **Verification:** The fan/motor should stop. The "RUN" lamp should extinguish.
5. While the motor is running in "HAND" mode, press the **Emergency Stop** button.
 - **Verification:** The motor must stop immediately. The LCP may show a "TRIP" or "FAULT" status.
6. **Testing and Commissioning:** Verifying motor rotation and initial performance.
7. Reset the E-Stop and any fault on the panel before proceeding. If Available

11.2 Status Indication

The LCP provides immediate, at-a-glance visual feedback on the equipment's status via indicator lamps on the panel door:

1. **RUN / ON Light (Typically Green):** Indicates that the motor is energized and running.
2. **STOP / OFF Light (Typically Red):** Indicates that the motor is stopped.
3. **TRIP / FAULT Light (Typically Amber/Yellow):** Indicates that the equipment has stopped due to a protective device tripping (e.g., motor overload, phase failure), alerting technicians to a problem.

Verify the fan's complete sequence of operation as per the design intent. This includes testing any interlocks (e.g., an exhaust fan must run whenever its corresponding fresh air fan is running) and any interface with the fire alarm system (e.g., shutdown on fire alarm).

12.0 Automatic System Activation Test (Fire Alarm Interface)

1. Preparation:

- Place the fan's LCP selector switch in the **"AUTO"** position.
- Station technicians with radios at the fan location, relevant damper locations, and test areas (e.g., staircase landings).
- Ensure all personnel are clear of the equipment.

2. Fire Alarm Activation:

- The Fire Alarm Contractor shall activate the system by triggering the appropriate detector (e.g., smoke detector in a lobby, manual call point) that is linked to the fan in the Cause & Effect Matrix.

3. **Verify System Response:** The following must occur **automatically and immediately** upon alarm activation:

- **Fan Starts:** The emergency fan must start automatically. The technician at the LCP must confirm this and that the "Run" light illuminates.
- **Dampers Operate:** Any associated motorized fire dampers must move to their designated fire-mode position (e.g., dampers on the fire floor open, others close). Technicians must visually confirm this.
- **Other Systems Interface:** Any linked systems (e.g., normal AC systems shutting down) must operate as per the Cause & Effect Matrix.

Step 3: Performance Verification in Fire Mode

(This is the most critical phase and is often witnessed by the RI/PE/SCDF Inspector)

- **For Staircase Pressurization Fans:**

1. **Measure Pressurization:** With the system running and all staircase doors closed, use a differential pressure meter to measure the pressure difference between the staircase and the adjacent lobby at multiple floors (top, middle, bottom).
 - **Acceptance Criterion:** The pressure must be within the SCDF Fire Code limits (e.g., typically between 25 Pa and 80 Pa).
2. **Measure Airflow Velocity:** Open one door on the fire floor and one door at the ground floor discharge. Use a vane anemometer to measure the air velocity through the open doorway.
 - **Acceptance Criterion:** The velocity must meet the minimum requirement in the Fire Code (e.g., >0.75 m/s).
3. **Measure Door Opening Force:** With the system at full pressure, use a door force gauge to measure the force required to open a staircase door.
 - **Acceptance Criterion:** The force must **not exceed** the maximum allowed by the Fire Code (e.g., <110 Newtons).

- **For Smoke Spill / Exhaust Fans:**

1. **Verify Airflow at Source:** Go to the designated smoke exhaust grille on the activated (fire) floor. Measure the airflow being extracted using an air capture hood or vane anemometer.

- **Acceptance Criterion:** The measured exhaust rate must meet or exceed the design value in the approved drawings.
- 2. **Verify Pressurization (if applicable):** Check that adjacent areas (e.g., lobbies) maintain a positive pressure relative to the smoke extraction zone.
- 3. **Smoke Visualization Test (Optional but recommended):** With the consent of the building owner and authorities, release a small amount of theatrical smoke in the fire zone. Visually confirm that the smoke is effectively captured and exhausted by the system and does not spread to adjacent protected areas.

Step 4: System Reset and Fault Indication Test

1. **Reset:** The Fire Alarm Contractor shall reset the main fire alarm panel.
2. **Verify Reset:** Confirm that the emergency fan stops and all motorized dampers return to their normal, non-fire position.
3. **Fault Simulation:** Manually trip the fan's motor overload relay at the LCP.
 - **Acceptance Criterion:** A "Fan Fault/Trip" signal must be registered at the main fire alarm panel and/or the Fire Command Centre. This test is mandatory.

13.0 Commissioning With BMS

1. **Point-to-Point Verification:** Verify all status points (Run, Fault/Trip) and command points (Start/Stop, Speed Control) from the LCP are correctly wired and appear on the BMS graphic.
2. **Remote Command Test:** Place the fan's LCP selector switch in "**AUTO**" mode.
 - Send a "Start" command from the BMS and verify the fan starts.
 - Send a "Stop" command and verify the fan stops.
3. **Status Feedback Test:**
 - While the fan is running, verify the "Run Status" on the BMS is "ON" or "Running."
 - Manually trip the motor's overload relay at the LCP. Verify the fan stops and the "Fault" or "Trip" status appears on the BMS. Reset the fault.

4. **VSD Speed Control:** If applicable, send a speed control signal (e.g., 50%) from the BMS and verify the VSD's output frequency changes accordingly.

13.1 Functional Testing

Perform a final functional test by running the system under its normal automated control sequence for to ensure stable and reliable operation without nuisance tripping.

14.0 Documentation and Handover

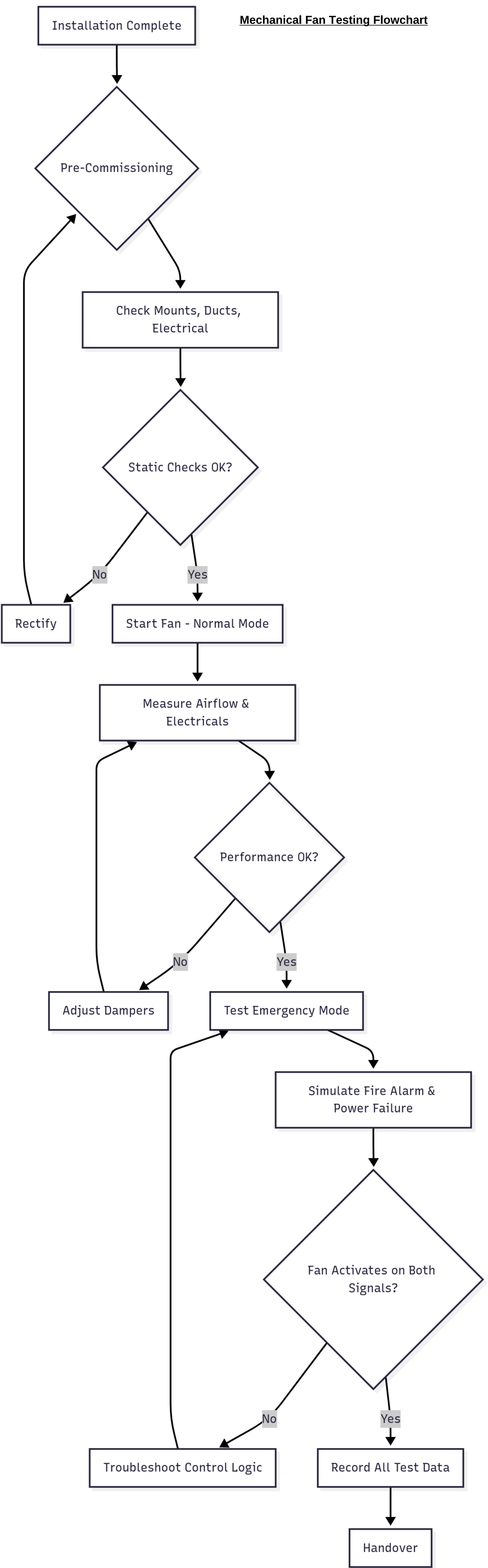
Throughout the process, record all readings, observations, and test results on the Commissioning Test Sheet for the system.

1. Calibration certificates for all testing equipment.
2. Completed test sheets for all equipment, including air balancing results.
3. Any failures, discrepancies, or non-conformities shall be recorded on a "Punch List" for rectification.
4. Upon successful completion of all tests, the Commissioning Engineer and Supervisor will sign the test sheet, confirming the System is safe and fully functional.
5. The completed and signed documentation will be included in the final Testing & Commissioning report for handover.
6. Compile the report for submission and verification with Maincon M&E / RTO or Consultant.
7. A formal handover of the commissioned system will be conducted with the client and facilities management team, including a demonstration of the system's operation.

15.0 Health and Safety

- All personnel must adhere strictly to the approved Risk Assessment and PTW.
- Mandatory PPE (safety glasses, gloves, safety footwear) shall be worn at all times.
- LOTO procedures are mandatory for any physical work or adjustment on the fan or inside the LCP.
- Ensure safe access for all work at height.
- Be aware of rotating equipment; never insert hands, tools, or probes into a running fan. Ensure all guards are in place.

APPENDIX



OPERATION DATA FOR MECHANICAL VENTILATION FAN			
PROJECT NAME			
LOCATION		EQUIPMENT DESIGNATION	
AREA SERVED		EQUIPMENT MODEL	
SYSTEM		MANUFACTURER	

S/No	DESCRIPTION	UNIT	TECHNICAL DATA
1	Air flow	CMH	
2	External static pressure	Pa	
3	Motor	kW/RPM	/
4	Type of fan	-	
5	Power supply	V/P/Hz	

S/No	DESCRIPTION	UNIT	MEASURED DATA
1	Total Fan air flow	CMH	
2	Thermal Overload Range / Setting (A)	A	/
3	Fan Starting Current	Amp	
4	Fan Running Current	Amp	
5	VSD Running Hz	Hz	
6	External static pressure	Pa	
7	Others		

Comments: (If there is no additional comments enter NONE)
--

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			

**Air Flow Testing & Commissioning Report**

Project Title :	Power Source (V/Ph/Hz):
Equipment Model:	FORMULA:(CMH = Average Velocity × (Duct or Grille Size) ×3600 CMH
Starting Ampere (A):	Running Ampere (A)
Location:	

No:	Equipment Designation	Area Served	Design CMH	Louver / Duct / Grille Size	P-1	P-2	P-3	P-4	P-5	Average Vel.M/s	Total CMH

Comments :-		
Tested By :	Witnessed By:	Verified By :
Signature :	Signature :	Signature :
Name :	Name :	Name :
Designation :	Designation :	Designation :
Company :	Company :	Company :
Date:	Date:	Date:



GRACECAL PTE LTD (GST Reg No: 201611630E)

31 Woodlands Close #02-05 Woodlands Horizon Singapore 737855
Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CLIENT : **WINNER ENGINEERING PTE LTD**
18 Woodlands Industrial Park E1
Singapore 757738

CERTIFICATE NO : GCC241442
ISSUE DATE : 06-SEP-2024

INSTRUMENT : **MULTI FUNCTION ANEMOMETER**
MAKE : TSI
MODEL NO. : TA465-P-NB
PART NO. : ---
SERIAL NO. : TA4652029012

WORKS ORDER NO : GWO241601
DATE RECEIVED : 02-SEP-2024

DATE CALIBRATED : 06-SEP-2024
DUE DATE : 06-SEP-2025

AMBIENT TEMPERATURE:
(23 ± 5) °C
RELATIVE HUMIDITY:
(55 ± 10) % r.h.

RANGE : -----
GRADUATION : -----
(TAG NO.) : -----
PAGE : 1 of 2
STATUS : AS FOUND

The described instrument has been calibrated at Laboratory under the ambient conditions stated above.

This certificate provides traceability of measurement to the International System of Units (SI) and/or to units of measurement realized at the National Metrology Centre (NMC), Singapore or other recognized national metrology institutes.

The Instrument was calibrated by comparison with laboratory calibration testing equipment.

1. Gas Deadweight Pressure Tester S/N: 558735

Calibration Method

Calibration Method is in accordance with ISO/IEC 17025. The calibration method was carried out according to Technical Calibration Procedure AS-TM-M03 as a guide.

Calibration Results

1. The results of calibration are given on the attached calibration data sheet(s).
2. The expanded uncertainty of measurement associated with the calibration is estimated at a confidence level of approximately 95% with a coverage factor of k=2.00.
3. The user should determine the suitability of the instrument for its intended use.

Calibrated By:

PREM

ID: 748656

Approved By:

KUMAR

ID: 748670





GRACECAL PTE LTD (GST Reg No: 201611630E)

31 Woodlands Close #02-05 Woodlands Horizon Singapore 737855
Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CERTIFICATE NO : GCC241442
ISSUED DATE : 06-SEP-2024

WORKS ORDER NO : GWO241601
PAGE : 2 of 2

APPLIED ACTUAL VALUE (Pa)		MEAN INSTRUMENT READING (Pa)				
		Ascending			Descending	
		Before Adj.	After Adj.	Error	Before Adj.	After Adj.
				Error		
Pressure Measurement						
0.00	0.1	---		-0.10	0.0	---
50.46	50.7	---		-0.24	50.7	---
477.53	477.8	---		-0.27	477.8	---
916.92	917.2	---		-0.28	917.2	---